Necessary horizontal ellipticity and the Rockland condition on Heisenberg groups

A strengthened partial result for arXiv:2007.14532

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Status. Candidate partial result, likely valid, pending human review. It proves a necessary principal-symbol condition on all stratified groups and maximal hypoellipticity on Heisenberg groups, but it does not settle the source's condition (ii).

1 Source question

Let G be a stratified homogeneous group with horizontal basis $X_1, \ldots, X_m$ and homogeneous dimension Q. Van Schaftingen and Yung consider homogeneous operators

$$A(D) = \sum_{\gamma \in \{1,\ldots,m\}^k} A^\gamma X_{\gamma_1} \cdots X_{\gamma_k} : C^\infty(G; V) \longrightarrow C^\infty(G; E)$$

and the endpoint estimate

$$\left\| D^{k-1} u \right\|_{L^{Q/(Q-1)}(G;V)} \leq C \left\| A(D) u \right\|_{L^1(G;E)}, \qquad u \in C_c^\infty(G; V). \tag{1}$$

They ask whether (1) forces (i) maximal hypoellipticity and (ii) a compatible operator with cocanceling symmetrization [1].

vector X_i, $i = 1, \ldots, m$, gives rise to a left-invariant vector field on G, which by abuse of notation we also denote by X_i. Let now k be a positive integer, and $\mathcal{I}_k$ be the index set $\{1, \ldots, m\}^k$. For $\gamma = (\gamma_1, \ldots, \gamma_k) \in \mathcal{I}_k$, we write X_γ for the left-invariant differential operator on G given by

$$X_\gamma := X_{\gamma_1} \ldots X_{\gamma_k}. \tag{1.6}$$

Note that X_γ depends on the ordering of the indices within γ if the group G is not abelian. If $A^\gamma \in \operatorname{End}(V; E)$ for all indices $\gamma \in \mathcal{I}_k$, then

$$A(D) := \sum_{\gamma \in \mathcal{I}_k} A^\gamma X_\gamma,$$

defines a homogeneous left-invariant linear partial differential operator $A(D)\colon C^\infty(G; V) \to C^\infty(G; E)$ of order k on G from V to E with real coefficients. We are interested in when the inequality

$$\|D^{k-1} u\|_{L^{Q/(Q-1)}(G;V)} \le C \|A(D) u\|_{L^1(G;E)}, \tag{1.7}$$

holds for such an operator $A(D)$, for all $u \in C^\infty_c(G; V)$, where $D^{k-1} u := (X_\gamma u)_{\gamma \in \mathcal{I}_{k-1}}$. The inequality (1.7) is the natural generalization of (1.2) to stratified homogeneous groups.

If $A(D) = D$, then (1.7) is an endpoint Sobolev inequality which is known to hold [18, 19]. In the particular case, where G is a Heisenberg group, Baldi, Franchi and Pansu have proved that (1.7) holds when $A(D)$ is a (first- or second-order) operator of the Rumin complex [1, 2]; their proof relies on the structure of the Rumin complex and on a

Figure 1: The endpoint estimate (1.7), source PDF page 2.

Bourgain–Brezis duality estimate on stratified homogeneous groups [14] generalizing the Euclidean results [34–36].

Our main result asserts that for a homogeneous left-invariant linear partial differential operator $A(D)\colon C^\infty(G; V) \to C^\infty(G; E)$ of order k as above, if

(i) $A(D)$ is maximally hypoelliptic, that is, if there exists some $C > 0$ such that

$$\|D^k u\|_{L^2(G;V)} \le C \|A(D) u\|_{L^2(G;E)}$$

for all $u \in C^\infty_c(G; V)$, and

(ii) there exists a finite dimensional inner product space F, a positive integer ℓ and some linear homogeneous left-invariant partial differential operator $L(D) = \sum_{\lambda \in \mathcal{I}_\ell} B^\lambda X_\lambda$ of order ℓ on G from E to F (so each $B^\lambda \in \operatorname{End}(E; F)$) such that the symmetrized operator $\operatorname{Sym}(L)(D)$ is cocanceling, that is

$$\bigcap_{\xi \in \mathbf{R}^m} \Big\{ e \in E \;:\; \sum_{\lambda \in \mathcal{I}_\ell} \xi^\lambda B^\lambda[e] = 0 \Big\} = \{0\},$$

then (1.7) holds for all $u \in C^\infty_c(G; V)$ (see Theorem 4.1 below). Under the above hypotheses (i) and (ii) on $A(D)$, we also obtain Hardy–Sobolev inequalities of the form

$$\left\| \frac{D^{k-\ell} u}{\|x\|^\ell} \right\|_{L^1(G;V)} \le C \|A(D) u\|_{L^1(G;E)} \tag{1.8}$$

where $\|x\|$ is a homogeneous norm on G and $\ell \in \{1, \ldots, \min\{k, Q-1\}\}$ (see Theorem 5.1(a)

Figure 2: Conditions (i)–(ii), source PDF page 3.

$$\left\| \frac{D^{k-\ell} u}{\|x\|^{\ell}} \right\|_{L^p(G;V)} \leq C_p \|D^k u\|_{L^p(G;V)},$$

which together with (1.9) implies (1.11).) The estimates (1.7) and (1.8) are then respectively a limiting case of (1.10) and (1.11) as $p \to 1^+$. It may be worth pointing out that the maximal hypoellipticity condition on $A(D)$ is actually equivalent (under our other assumptions on $A(D)$) to a hypoellipticity condition on $A^t(D)A(D)$; see again Theorem 3.3. It is an interesting open question whether the validity of (1.7) for all $u \in C^\infty(G;V)$ implies the conditions (i) and (ii) on $A(D)$ on a general stratified homogeneous group G.

In order to apply our main theorems, given the operator $A(D)$ one must construct a compatible $L(D)$ as in condition (ii) above. In Proposition 6.1 we develop a robust way of doing so that works in many examples of interest. In particular, in Proposition 7.2 below, we obtain the Gagliardo–Nirenberg–Sobolev type estimate

$$\|D^{k-1} u\|_{L^{Q/(Q-1)}(G)} \leq C \sum_{i=1}^{m} \|X_i^k u\|_{L^1(G)}$$

Figure 3: The open question, source PDF page 4.

2 Proof intuition

There are two tests. First, if the horizontal commutative symbol has a kernel at a direction c, keep a test function compact along c while dilating all transverse coordinates. The only term with all k derivatives on the one-dimensional factor is killed by the symbol kernel. Every other term costs R^{-1}, leaving the same strict $R^{1/Q}$ contradiction as in the scalar first-order case.

Second, on a Heisenberg group, a failure of the Rockland condition occurs either at a character (already detected by the first test) or in a Schrödinger representation. A smooth representation-kernel vector gives an exact A-harmonic matrix coefficient, Schwartz horizontally and oscillatory in the center. Cut it off only at central scale R^2. Its left norm grows as $R^{2(Q-1)/Q}$, whereas every term on the right differentiates the cutoff, costs R^{-2}, and is integrated over an interval of length $O(R^2)$.

3 A necessary horizontal-symbol condition in all steps

Define the commutative horizontal symbol

$$A_{\mathrm{h}}(\xi) = \sum_{\gamma\in\{1,\dots,m\}^k} \xi_{\gamma_1}\cdots\xi_{\gamma_k} A^\gamma, \qquad \xi\in\mathbb{R}^m. \tag{2}$$

Theorem 1 (Horizontal injective ellipticity)**.** *If* (1) *holds on a stratified homogeneous group with* $Q\ge 2$*, then* $A_{\mathrm{h}}(\xi): V\to E$ *is injective for every* $\xi\in\mathbb{R}^m\setminus\{0\}$*.*

Proof. Suppose $A_{\mathrm{h}}(c)v=0$ for nonzero $c\in\mathbb{R}^m$ and $0\neq v\in V$. Choose the first horizontal basis vector in the direction c. In the new coordinates, the coefficient of X_1^k kills v.

Complete the horizontal basis to a homogeneous basis $Y_1,\dots,Y_N$ of $\mathfrak{g}$, with weights $w_1=\cdots=w_m=1$ and $\sum_{\nu=1}^N w_\nu = Q$. In exponential coordinates of the first kind,

$$X_j = \partial_{x_j} + \sum_{\nu>m} p_{j\nu}(x)\partial_{x_\nu}, \tag{3}$$

where $p_{j\nu}$ is weighted-homogeneous of degree $w_\nu - 1$. Choose $\phi\in C_c^\infty(\mathbb{R})$ with $\phi^{(k-1)}\neq 0$ and nonzero $\eta\in C_c^\infty(\mathbb{R}^{N-1})$, and set

$$\eta_R = \eta(R^{-w_2}x_2,\dots,R^{-w_N}x_N), \qquad u_R = \phi(x_1)\eta_R v.$$

The monomial calculation from (3) shows that every fixed word of horizontal derivatives in which at least one derivative falls on η_R is $O(R^{-1})$ on the support. Indeed, a monomial x^α in $p_{j\nu}$ contributes at most

$$R^{\sum_{\mu\ge 2} w_\mu\alpha_\mu} R^{-w_\nu} = R^{-1-\alpha_1} \le R^{-1},$$

and differentiating such polynomial coefficients a fixed number of times does not worsen this bound.

Consequently the X_1^{k-1} component of $D^{k-1}u_R$ is

$$\phi^{(k-1)}(x_1)\eta_R v + O(R^{-1}),$$

so, with $q=Q/(Q-1)$,

$$\left\| D^{k-1}u_R \right\|_{L^q} \ge cR^{(Q-1)/q} = cR^{Q-2+1/Q} \tag{4}$$

for large R. In $A(D)u_R$, the only term with every derivative on ϕ is $A_{\mathrm{h}}(c)v\,\phi^{(k)}\eta_R = 0$; every other term is $O(R^{-1})$. Since the transverse support has measure $O(R^{Q-1})$,

$$\|A(D)u_R\|_{L^1} \le CR^{Q-2}. \tag{5}$$

Equations (4)–(5) contradict (1) as $R\to\infty$. $\square$

4 The full Rockland condition on Heisenberg groups

Let $\mathbb{H}^n$ have coordinates $(z,t) \in \mathbb{R}^{2n} \times \mathbb{R}$, horizontal fields $X_1, \ldots, X_{2n}$, central weight 2, and homogeneous dimension $Q = 2n+2$.

Theorem 2 (Maximal hypoellipticity on $\mathbb{H}^n$)**.** *For an arbitrary homogeneous $A(D) : C^\infty(\mathbb{H}^n; V) \to C^\infty(\mathbb{H}^n; E)$ of order k, validity of* (1) *implies that $A(D)$ satisfies the Rockland condition. Equivalently, $A(D)$ is maximally hypoelliptic.*

Proof. Suppose the Rockland condition fails. Then some nontrivial irreducible unitary representation π has a nonzero smooth vector $w \in \mathcal{H}_\pi^\infty \otimes V$ such that

$$d\pi(A)w = 0. \tag{6}$$

For a one-dimensional representation, $d\pi(A)$ is a nonzero scalar multiple of $A_{\mathrm{h}}(\xi)$ for a nonzero horizontal covector ξ, contradicting Theorem 1. We may therefore take $\pi = \pi_\lambda$ to be a Schrödinger representation, $\lambda \neq 0$.

Its smooth vectors are $\mathcal{S}(\mathbb{R}^n)$-vectors. Choose $\zeta \in \mathcal{S}(\mathbb{R}^n)$ and form the V-valued matrix coefficient

$$F(g) = \langle \pi_\lambda(g)w, \zeta \rangle_{\mathcal{H}_\pi}. \tag{7}$$

Here ζ is chosen so that some horizontal derivative of order $k-1$ of F is nonzero. Such a choice exists: if all words of length $k-1$ killed w, then in particular $d\pi(X_j)^{k-1}w = 0$ for every j. For $k > 1$ the skew-adjointness of $d\pi(X_j)$ gives $d\pi(X_j)w = 0$ for every j; for $k = 1$ the claim is immediate from $w \neq 0$. Since the horizontal fields generate the Lie algebra, the former alternative would make w invariant in a nontrivial irreducible representation, a contradiction.

With the source convention $Xf(g) = \frac{d}{ds}|_{s=0} f(g \exp(sX))$, (6) gives

$$A(D)F = 0. \tag{8}$$

The explicit Schrödinger formula shows

$$F(z,t) = e^{i\lambda t} f(z),$$

where f and all horizontal derivatives of the coefficient are Schwartz in z.

Choose nonzero $\chi \in C_c^\infty(\mathbb{R})$ and put

$$U_R(z,t) = F(z,t)\chi(t/R^2).$$

These functions are not yet compactly supported in z, but multiplication by $\rho(z/S)$ and passage $S \to \infty$ makes them admissible in (1); all required convergence follows from Schwartz decay.

Let γ be a word of length $k-1$ with $X_\gamma F \neq 0$. Product differentiation gives

$$X_\gamma U_R = \chi(t/R^2) X_\gamma F + \mathcal{E}_R.$$

Every term in $\mathcal{E}_R$ differentiates the central cutoff at least once. Since a horizontal Heisenberg field has the form $\partial_z +$ a linear function of z times ∂_t, such a term is R^{-2} times a Schwartz function of z and a bounded derivative of $\chi(t/R^2)$ (terms with more cutoff derivatives are smaller). Hence

$$\left\| D^{k-1} U_R \right\|_{L^q} \geq cR^{2/q} - O(R^{2/q-2}) \geq c' R^{2/q}. \tag{9}$$

By (8), every product-rule term in $A(D)U_R$ also has at least one derivative on the cutoff. Its z factor is Schwartz and its central support has length $O(R^2)$, so the factor R^{-2} cancels that length:

$$\|A(D)U_R\|_{L^1} \leq C \tag{10}$$

uniformly in R. Equations (9)–(10) contradict (1).

Thus A is Rockland. The equivalence with maximal hypoellipticity follows from the Rockland theorem applied to the formally positive operator $A^t(D)A(D)$, as in Theorem 3.3 of [1]. $\square$

5 What remains open

Theorem 1 supplies a necessary shadow of condition (i) on every stratified group, and Theorem 2 proves condition (i) itself on every Heisenberg group. Neither theorem constructs the compatible cocanceling operator required in condition (ii). Moreover, extending the matrix-coefficient cutoff from Heisenberg groups to arbitrary higher-step groups requires uniform control in general induced-representation models. The original question therefore remains open beyond the stated conclusions.

6 Verification and novelty status

The proof uses no computation. Its main checks are: the strict exponent gap $1/Q$ in Theorem 1; the left/right infinitesimal-representation convention in (7); and the R^{-2} central-cutoff cost in Theorem 2.

A bounded search through the run indexes, local arXiv corpus, exact-title queries, and the source's three citing works found no explicit statement of these strengthened necessity results. Novelty confidence is moderate rather than definitive, and human expert review is recommended.

References

[1] J. Van Schaftingen and P.-L. Yung, *Limiting Sobolev and Hardy inequalities on stratified homogeneous groups*, Ann. Fenn. Math. 47 (2022), no. 2, 1065–1098; arXiv:2007.14532; DOI: 10.54330/afm.120959.